May the Force Be With You

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Static Electricity

Field(s) of Interest: Static Electricity, Physics, Physical Science, Materials Science

Brief Overview (1-3 sentences):
Mentees will explore static electricity through hands-on activities using balloons, tissue paper, aluminum foil, and soda cans. By the end of the lesson, they should understand how electric charges form, transfer, and interact through attraction and repulsion.

Agenda:

- Introduction (5 min)
- Module 1: Let's Boogie (15 min)
- Module 2: A Shocking Attraction (20 min)
- Module 3: Can you feel the force? (15 min)
- Conclusion (5 min)

Main Teaching Goals/Key Terms:

- ➔ Static electricity
- ➔ Friction
- ➔ Charge transfer
- ➔ Electrons & protons
- ➔ Neutral objects
- ➔ Polarization
- ➔ Conductors & insulators
- ➔ Electric field
- ➔ Charge redistribution
- ➔ Electric force
- ➔ Attraction & repulsion
- ➔ Induced charge

Background for Mentors

Module 1

- Static electricity
- Friction
- Charge transfer
  - Electrons
  - Protons
- Neutral objects

Static electricity is the *buildup of electric charge on the surface of an object*. It occurs when certain materials rub against each other and electrons transfer from one material to another. This process is called **charge transfer**, and the rubbing creates friction between the materials.

Friction is the *force that occurs when two surfaces rub against each other*. When certain materials rub together, electrons can transfer from one material to another. This leaves one object negatively charged (gains electrons) and the other positively charged (loses electrons).

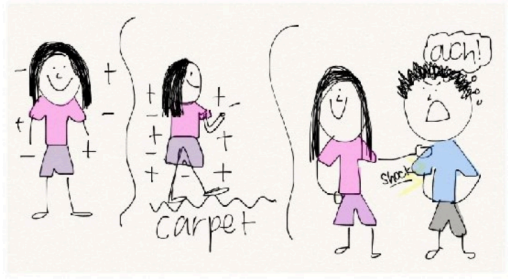


Figure 1. *The transfer of electric charge through friction.*

For example, when mentees rub a balloon against their hair, electrons can transfer from their hair to the balloon, leaving the balloon negatively charged and their hair relatively positively charged. This is why hair stands up and gets attracted to the balloon!

Electrons are *tiny negatively charged particles* that can move between materials, while **protons** are *tiny positively charged particles* located inside an atom's nucleus. Unlike electrons, protons generally do not move between objects during these activities. An object that gains electrons becomes negatively charged, while an object that loses electrons becomes positively charged. An *object with equal amounts of positive and negative charge* is considered **neutral**.

Module 2

- Charge redistribution
- Polarization
- Conductor
- Insulator
- Electric Field

Charge Redistribution *occurs when electric charges within an object rearrange when a charged object is brought nearby*. Positive and negative charges shift to different areas, creating an uneven distribution of charge. **Polarization** *is the result of charge redistribution, where positive and negative charges become separated within a neutral object*. The object does not gain or lose electrons overall, but this separation allows it to be attracted to a nearby charged object.

Materials differ in how easily electric charges can move through them. A **conductor** is a *material that allows electric charges, particularly electrons, to move easily*. Metals, such as copper or aluminum foil, are good conductors because they contain electrons that can move relatively freely throughout the material. An **insulator** *does not allow electric charges to move easily*. Materials such as foam, rubber, and plastic are common examples because their electrons are more tightly bound.

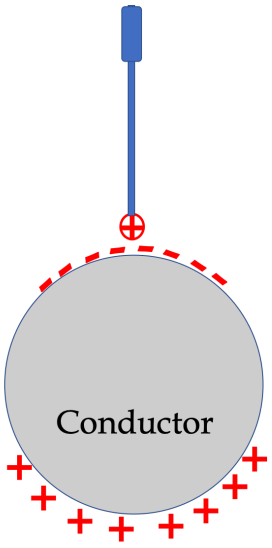


Figure 1. *Polarization of charges on a conductor when a charged object is introduced.*

For example, in Module 2's activity, the foam balls act as insulators because their electrons are tightly bound, while the aluminum foil acts as a conductor because its electrons can move freely. When a charged balloon is brought nearby, its electric field causes the charges in the foil to redistribute, creating polarization. The opposite charges are attracted to the balloon, causing the foam balls to move toward it. This demonstrates how conductors allow

	charge redistribution, while insulators restrict the movement of electric charges. An **electric field** is *the area around a charged object where it can exert an electric force on other charged objects*. This means that objects do not need to physically touch for an electric force to act on them. The strength of an electric field generally decreases as the distance from the charged object increases.

Module 3

- Electric force
- Attraction
- Repulsion
- Induced charge

An **electric force** is *a push or pull between objects caused by their electric charges*. Electric forces can act over a distance, meaning objects do not need to touch for them to attract or repel each other. There are two types of electric interactions: attraction and repulsion. **Attraction** *occurs when objects pull toward each other*, while **repulsion** *occurs when objects push away from each other*. Opposite charges attract, like charges repel.

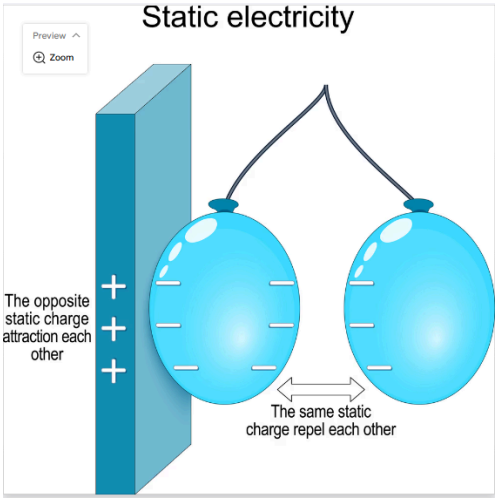


Figure 2. *Objects with similar charge repel, while objects with opposite charge attract.*

A simple way to demonstrate this to the mentees is by comparing electric charges to magnets. Opposite magnetic poles attract each other, while matching poles repel. Mentees can also think of attraction as being drawn toward their favorite food and repulsion as wanting to move away from a food they dislike. Although these are different types of interactions, they help explain the general idea of pulling toward or pushing away from something.

Induced charge is *the development of an induced positive charge on one side of an object and an induced negative charge on the other*. This occurs when charge redistribution causes polarization within a neutral object, separating its existing charges without changing its overall neutrality. These charges are not newly created; they are the result of existing charges shifting within the object, leaving it neutral overall.

This process happens during the soda can activity. The aluminum soda can is initially neutral, meaning it has equal amounts of positive and negative charge. When a negatively charged balloon approaches, electrons within the metal can are repelled and move toward the opposite side of the can. This

	leaves the side closest to the balloon relatively more positive. Because opposite charges attract, the can experiences a net force toward the balloon and begins to roll.

Introduction

Static electricity is something mentees experience in everyday life, even if they don't realize the science behind it. Have they ever gotten shocked after going down a slide, had their hair stand up after rubbing a balloon on it, or seen a balloon stick to a wall? These are all examples of static electricity!

Concepts to Introduce	**Questions to Pique Interest**
● **Static electricity:** explain that electricity doesn't always need wires or outlets. Sometimes electric charges can build up on objects and create forces ○ Analogy: have mentees think about getting shocked after sliding down a metal slide ● **Friction and charge transfer:** rubbing two materials together can cause electrons to move from one material to another ● **Positive and negative charges:** objects can have positive or negative charges depending on whether they lose or gain electrons ● **Attraction and repulsion:** electric charges can pull objects together or push them apart without touching	● Have you ever gotten shocked after going down a slide and a metal railing/someone? ○ What do you think caused the shock? ● What happens when you rub a balloon against your hair and then hold it near your head? ● Why do you think some objects are easier to move with static electricity than others? ● How do magnets work?
Scientists, Current and Past Events	**Careers and Applications**
● Charles-Augustin de Coulomb: A French physicist who studied the forces between electrically charged objects. His work led to Coulomb's Law, which describes how electric forces depend on the amount of charge and the distance between objects. ● Benjamin Franklin: Known for his famous kite experiment, Franklin helped demonstrate the connection between lightning and electricity. He also introduced the positive and negative naming convention for electric charges.	● Electrical engineer: designs electrical systems and devices that use electricity and electric fields ● Electronics engineer: works with electrical components and circuits found in phones, computers, and other electronics ● Material scientist: studies how different materials behave and how their properties can be used to develop new technologies

Module 1: Let's Boogie

Mentees create their own tissue paper people and use a charged balloon to make their figures dance! By rubbing the balloon against hair or fabric, mentees build up static electricity and observe how electric charges can attract lightweight objects without direct contact.

Teaching Goals	**Materials**
1. **Static Electricity:** a build up of electric charge on an object 2. **Friction:** the force created when two surfaces rub against each other 3. **Charge Transfer:** movement of electrons from one material to another a. **Electrons:** tiny negatively charged particles b. **Protons:** tiny positively charged particles 4. **Neutral objects:** objects with equal amounts of positive and negative charge	• 1 balloon per 3-4 group of mentees • 1 pair of scissors per 3-4 group of mentees • 1 tissue paper per mentee • 1 piece of paper per mentee • Markers • Tape • 2 pieces of fabric per site

Different Methods for Teaching

1. **Static electricity:** Ask mentees if they have ever been shocked after going down a slide or touching a doorknob.
2. **Friction:** Have mentees rub their hands together and think about how friction creates warmth. Explain that rubbing certain materials together can also cause electrons to transfer
3. **Charge Transfer:** Explain that hair is generous with electrons, while balloons are "hungry" for electrons. Have mentees rub a balloon against their hair, allowing electrons to transfer from the hair to the balloon.

Procedure

1. Place sheet of paper under the tissue paper to prevent markers from staining the table
    a. **Sites with younger mentees:** Make an outline of a figure on one tissue paper, stack multiple tissue papers under it and cut the stack so you have multiple figures
2. Have mentees draw out a 2D figure of a person on tissue paper, leaving extra space at the bottom of the feet so they can be taped to



Figure 1: *for younger sites stack tissue paper, make one outline and cut*

the table

3. Allow mentees to decorate their figures with markers
4. Have mentees carefully cut out their figures with scissors
5. Tape the feet of the figure's feet to the table
6. Using the balloon, charge it by rubbing it on hair or an object that has high electron affinity (cotton shirt, carpet)
7. Hover the balloon over the tissue paper figures and watch how they dance!



Figure 2: *you can make multiple at once for mentees who are too young to handle scissors*



Figure 3: *Draw figure on tissue paper and cut it out*



Figure 4: *Tape feet of figure onto the table, charge the balloon, and hover balloon over figure*

Module 2: A Shocking Attraction

Mentees will work in groups to explore how charged objects interact with neutral objects through polarization. They will observe a mentor demonstration of a charged balloon bending a stream of water, then cover foam balls with aluminum foil and use charged balloons to move them between plates. These activities demonstrate how electric charges can redistribute within objects and create attraction.

Teaching Goals

1) **Charge redistribution:** rearrangement of electric charges within an object
2) **Polarization:** the result of charge redistribution, where positive and negative charges become separated within a neutral object.
3) **Conductor:** a material that allows electric charges to move easily
4) **Insulator:** a material that does not allow electric charges to move easily
5) **Electric Field:** area around a charged object where it can exert an electric force

Materials

Demo:

- 1 balloon
- 1 measuring cup
- 1 paper cup
- 1 food coloring
- Access to water/bring a water bottle

Activity:

- 5 foam balls per 4-5 mentees
- 1 sheet of aluminium foil
- 1 balloon per 3-4 mentees
- 1 paper plate per 3-4 mentes
- 1 paper cup/bowl per 3-4 mentees
- 2 pieces of fabric per site

Different Methods for Teaching

1) **Charge redistribution:** Explain to mentees that when an ice cream truck arrives at a park, kids move toward the side where the truck is parked. Similarly, a charged object causes charges within a nearby neutral object to shift toward or away from it.
2) **Conductor:** Have mentees think about a crowded hallway between classes. What happens when students can move freely from one end to the other? Conductors work similarly because they allow electrons to move easily through the material.
3) **Insulator:** Have mentees think about a locked classroom door. Even if students want to move to the other side, the door prevents them from passing through. Insulators make it difficult for electrons to move freely through a material.

Preparing the demo:

1) Grab a balloon, measuring cup, paper cup, and food coloring
2) Have one mentor fill the measuring cup with water
3) Add food coloring so mentees could see the stream move with better visibility
4) Show the demo to each group of mentees **WHILE** they are covering their foam balls with aluminum foil

Demo: Bend the water

1) Have one mentor rub the balloon against their hair or a piece of fabric to create static charge
2) Have another mentor carefully pour a very thin, steady stream of water into the paper cup
3) Slowly bring the charged balloon near the stream of water without touching it.
4) Have mentees observe how the stream of water bends toward the balloon.
5) Have mentees guess what happens when they move the balloon closer or farther away
6) Discuss with mentees why they think the water moved even though the balloon never touched it

Activity:

1) Pair mentees into a group of 4-5
2) Give each group 5 foam balls, a sheet of aluminum foil, 1 paper plate, 1 paper bowl/cup, and a balloon
3) Have mentees wrap the foam balls with aluminum foil
    a) Make sure that the layer of aluminum foil on the ball is thin
    b) Too thick = too heavy
4) Instruct mentees to charge the balloon by rubbing it against hair or a piece of fabric
5) Hold the charged balloon above the foam balls and observe how the balls move toward the balloon
6) **Challenge:**

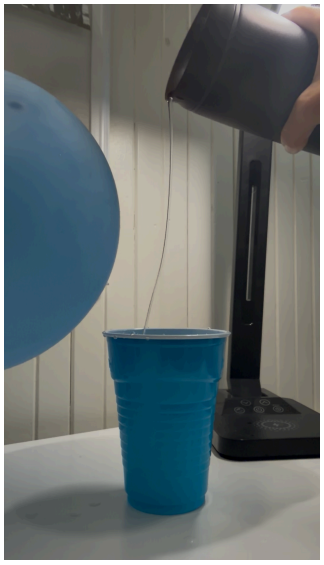


Figure 1: *pour water steadily using the beaker then charge balloon, water should bend towards the balloon*



Figure 2: *YOU DO NOT NEED TO COVER PLATE IN ALUMINUM FOIL, this is just to demonstrate the activity*

a) Have mentees compete to see who could move the foam balls from one paper plate to a paper bowl/cup the fastest	 **Figure 3:** *example of correct foil coverage on the left vs incorrect foil coverage on the right*

Module 3: Can you feel the force?

Mentees will work in groups to use charged balloons to move empty soda cans without touching them. This activity demonstrates how electric forces can attract neutral objects through induced charges.

Teaching Goals	**Materials**
1) **Electric force:** a push or pull between objects caused by their electric charges 2) **Attraction:** to pull objects toward each other 3) **Repulsion**: to push objects away from each other 4) **Induced charge:** the development of an induced positive charge on one side and an induced negative charge on the other.	• 1 balloon per 3-4 mentees • 1 empty soda can per 4 mentees • 2 pieces of fabric per site

Different Methods for Teaching

1) **Electric force:** Think about pushing a friend on a swing. The harder you push, the more the swing moves! Similarly, electric force is the push or pull between objects caused by their electric charges.

2) **Attraction:** Have mentees think about their favorite food. When they smell it cooking, they naturally want to move closer to it! Similarly, opposite electric charges attract each other.

3) **Repulsion:** Have mentees think about a food they really dislike. If someone offered it to them, they would probably deny the offer. Also repulsion sounds like repulse which means dislike in a sense.

4) For all of these teaching goals, magnets are a good example (especially for older mentees). One end of a magnet has a positive charge and the other end has a negative charge.
   a) When you hold the positive end of one magnet and the negative end of another near each other, you can feel them pull towards each other. This is attraction.
   b) When you hold the two positive ends or two negative ends of two magnets near each other, you can feel them push away from each other. This is repulsion.
   c) Both of these feelings are due to the electric force caused by the magnet's charge.

Procedure

1) Have mentees charge the balloon
2) Place an empty soda can on its side and have mentees hold the balloon near the can
3) Mentees will observe how a charged balloon can create a force that moves a soda can toward the balloon without direct contact
4) Have mentees compete by guiding cans in a straight line across the classroom.
    a) For classrooms with carpet or tile floors, line a couple rows of tables together and have mentees compete with each other, guiding their can from one side to another
5) Discuss with mentees what caused the cans to move



Figure 1: *the negative charges on the balloon attracts the neutral soda can*

Classroom Notes

The activity for module 3 works best on **smooth surfaces** because there is less friction between the can and the ground.

If the can does not move, have mentees recharge their balloons by rubbing them against hair or fabric.

Encourage mentees to experiment rubbing their hair with different fabric materials.

Conclusion

After the lesson is finished have mentees reflect on what they discovered about static electricity. How can electric charges make objects move or attract each other without physically touching?

References

- Module 1 Activity: https://maglabweb.magnet.fsu.edu/magnet-academy/try-this-at-home/groovin-ghost/

Summary Materials Table

Material	**Amount per Site**	**Expected $$**	**Vendor (or online link)**
Balloons	1 per 3-4 mentees	$17.97	Amazon
Tissue paper (5 x 5 inches)	1 per mentee	$8.99	Amazon
Blank paper	1 per mentee	N/A	Hearst Cages
Markers	1 per 2-3 mentees	N/A	Hearst Cages
Scissors	1 per 3-4 mentees	N/A	Hearst Cages
Masking tape	2-3 per site	N/A	Hearst Cages
Measuring cups	1 per site	$23.99	Amazon
Food coloring	1 per site	N/A	Hearst Cages
Paper cups	1 per site	N/A	Hearst Cages
Paper bowls/cups	1 per 3-4 mentees	N/A	Hearst Cages
Paper plates	1 per 3-4 mentees	N/A	Hearst Cages
Foam balls	5 per 4-5 mentees	$35.98	Amazon
Aluminum foil	1 per site (large size for foam balls)	$17.91	Amazon
Soda cans	1 per 4 mentees	$29.95	Amazon